Trekking Management App Dev Project Report

1. Student Details

Name: Abhay Shukla
Roll Number: 24f3002105
Email: 24f3002105@ds.study.iitm.ac.in
About Me: I am pursuing my DATA SCIENCE AND APPLICATION . I have completed python programming and SQL and DSA courses from IIT Madras and other trusted platforms which has given me the latest skills and knowledge in these types of programming languages.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

To build a **Trekking Management Application** web application that allows **Admin, Trek Staff, and Users (Trekkers)** to interact with the system based on their roles, where user or trekker can book treks based on their interest and staff are assigned to guide trekkers and admin to look over all the things.

Approach:

The project is implemented using the Flask framework for the backend and SQL Alchemy for the ORM. The frontend is created using HTML and CSS for providing a user-friendly interface for booking and finding best treks. The approach includes creating models for Users, Trek and Booking, setting up appropriate relationships using python functions and operations. HTML and CSS used for easy and proper login and register page for user, staff and admin where admin is predefined. Admin dashboard facilitates all processes like add,edit and delete treks including approve,remove and blacklist both user and staff.

3. AI/LLM Declaration

I have created a very simple web application with no more complex features, with a very simple user interface and logics as taught in MAD 1 project sessions. Since I am a beginner in web development, I used **ChatGPT** to learn some concepts ,functions and SQL Alchemy terms to complete my project.

The extent of **ChatGPT** usage is around 10% for some external things like image insertion code suggestions(admin can insert image of treks) and also to apply some Jinja2 concepts.

4. Technologies and Frameworks Used

- **Flask:** Flask is a lightweight web framework that makes it simple to build web applications. It has minimal external dependencies, making it easy to work with and maintain.
- **Jinja2:** Jinja2 is an extensible templating engine for rendering dynamic HTML pages.
- **Flask-SQLAlchemy:** Flask-SQLAlchemy is a wrapper around SQLAlchemy specifically designed to integrate with Flask applications. It is a powerful Python Object-Relational Mapper (ORM) and database toolkit.
- **CSS: CSS**, which stands for **Cascading Style Sheets**, is the standard design language used to style and format the presentation of web pages.
- **HTML:** The standard markup language used to create and structure the framework of all web pages on the internet.

- **Werkzeug:** A foundational Python library that provides a comprehensive set of utilities. It handles core web mechanics like HTTP requests, URL routing, and interactive debugging.

5. Database Schema / ER Diagram

1. **User Table (Stores User Profile Details)**
   Fields:

- **id:** Primary key, unique identifier for each user.
- **name:** User's name, not nullable.
- **email:** User's email must be unique and not nullable.
- **password:** User's password, not nullable.
- **role:** User's role, not nullable.
- **contact:** User's contact.
- **approved:** boolean value, default is true.
- **blacklisted:** boolean value, default is false.

2. **Staff Profile (Stores Staff Profile Details)**
   Fields:

- **id:** Primary key, unique identifier for each staff member.
- **user_id:** Foreign key referencing the User table.
- **contact:** contact number of staff.
- **status:** status of staff, default is pending.

3. **Trek Table (Stores all information about treks)**
   Fields:

- **id:** Primary key, unique identifier for each trek.
- **name:** Trek's name, not nullable.
- **location:** Trek's Location, not nullable.
- **difficulty:** Trek's Difficulty, not nullable.
- **duration:** Trek's Duration, not nullable.
- **available_slots:** Integer value, not nullable.
- **assigned_staff_id:** Foreign Key referencing the User table.
- **status:** Trek's Status, default is open.
- **start_date:** Trek's Start Date.
- **end_date:** Trek's End Date.
- **description:** Information about Trek.
- **image:** Trek's Image.

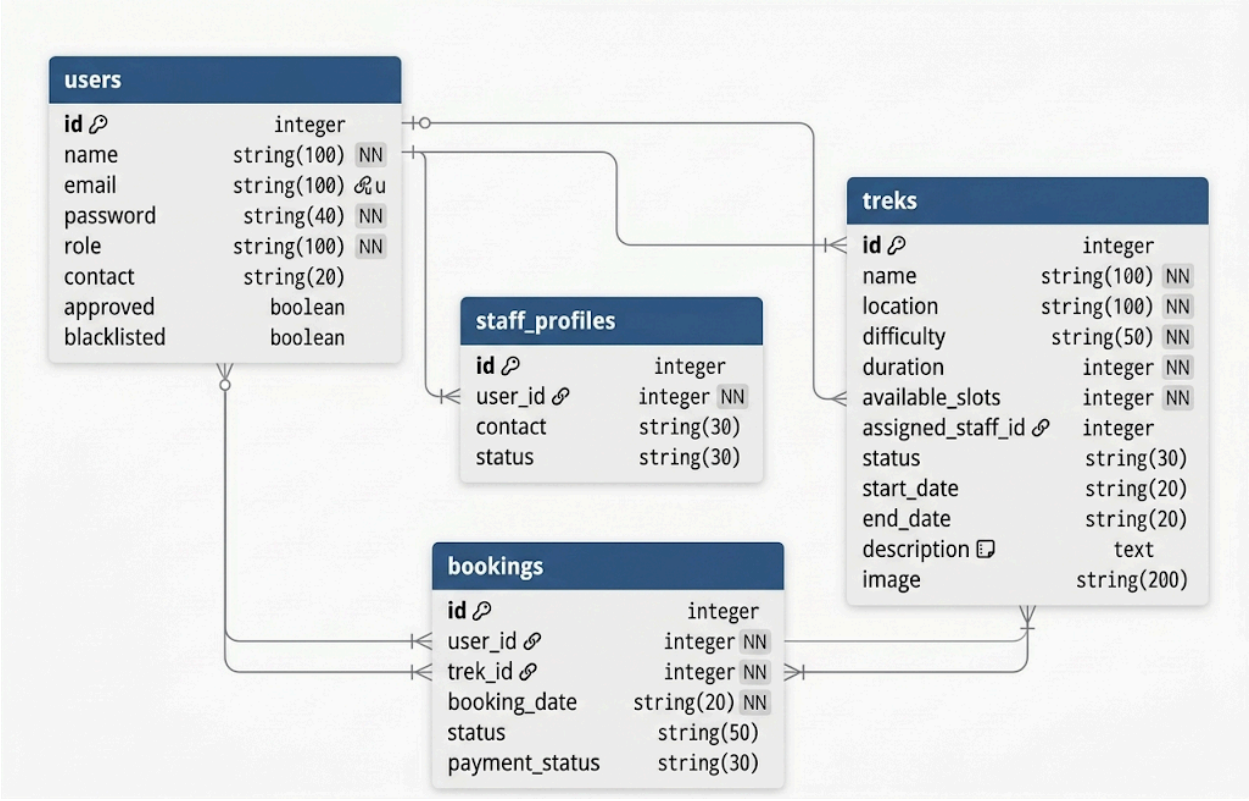
4. **Booking Table (Stores records of bookings)**
   Fields:

- **id:** Primary Key, unique identifier for each bookings.
- **user_id:** Foreign Key referencing the User Table, not nullable.
- **trek_id:** Foreign Key referencing the Trek Table, not nullable.
- **booking_date:** Booking Date, not nullable.
- **status:** status of booking, default is booked.
- **payment_status:** payment_status of booking, default is paid.

Relationships:

- **One-to-Many → User → Booking**
- **One-to-Many → User → Trek**
- **One-to-Many → Trek → Booking**

ER Diagram



Project video link: [click here to watch video]